Project Planning

Phase 4: Project Planning Phase

1. Planning Objective

The plan organizes work into stages so that the main journey can be built and tested before extras are attempted. The highest priority is a complete vertical slice: sign in, enter preferences, generate an itinerary, see the budget, save the trip and reopen it. Additional features are added only after this path works reliably.

2. Work Breakdown

Requirements and scope

Define the target user, the problem, the wizard fields and the itinerary structure. Confirm that booking, payments and verified live data are not part of the first version.

Environment, scaffolding and responsive interface

Set up Git, Python 3.11 and Node.js; scaffold the Vite and React frontend with Tailwind CSS; build the pages (landing, login, sign-up, dashboard, wizard, itinerary, saved trips, edit, profile, settings, not-found) and verify a clean production build.

Database schema and security

Write the SQL migration for profiles, trips, trip preferences, itinerary days and activities, add timestamps and signup triggers, and enable row-level security on every table.

Backend and API

Implement Pydantic schemas, the bearer-token guard, health, trips and export routes with ownership checks, and the budget service.

AI itinerary engine

Design the structured JSON prompt, connect Gemini and OpenAI, add the heuristic fallback, and implement day regeneration.

Integration and budget logic

Connect the Axios client with automatic bearer-token injection, add the live budget summary and variance alerts, and support inline editing with cost recalculation.

Export, testing and verification

Add the print stylesheet and the printable HTML export, write backend and frontend tests, and prepare a manual end-to-end checklist.

Deployment and documentation

Prepare vercel.json, render.yaml and the Procfile, deploy the frontend, write the setup guides, capture screenshots and complete the documentation phases.

3. Milestones

1. Scope and requirements approved.
2. Frontend scaffold and responsive pages built; production build verified.
3. Supabase schema and row-level security written.
4. FastAPI backend with authenticated trip endpoints.
5. AI itinerary engine with structured output and fallback.
6. Frontend and backend integrated with the budget calculator.
7. Print/PDF export, automated tests and manual checklist prepared.
8. Frontend deployed on Vercel; backend deployment configuration prepared.
9. Screenshots, test evidence and the eight documentation phases completed; demonstration recorded.

4. Risk Management

Risk	Impact	Mitigation
AI provider quota, rate limit or network failure	Generation fails	Fallback to the heuristic generator and the client-side generator; clear error toast
Missing or wrong Supabase configuration	Sign-in and persistence unavailable	Demo mode with localStorage; verify environment variables in each environment
Misleading travel information	User relies on wrong data	Label costs as estimates; show the budget warnings; require user verification
CORS mismatch between Vercel and Render	Frontend cannot reach the API	Add the Vercel origin to CORS_ORIGINS and test /health
Incomplete evidence at submission	Weak documentation	Reserve time for screenshots, test logs and demo; check that no secrets are visible

5. Completion Criteria

The project is ready for final review when the core flow works in the deployed environment, limitations are documented, screenshots are inserted with captions, test results reflect actual runs, and the demonstration link opens for reviewers.